

VOCAL MIC SOURCE Technical Manual

Vocal, microphone, and source-tone shaping

Build or protect the source before downstream color, dynamics, and spatial processing.

Included products: GOLD, SILVER, WPR, Aegis Nova, TRIPLER, EchoLocate

Routing authority: Lead vocal: GOLD -> AEGIS NOVA -> SILVER. Background vocal: GOLD -> TRIPLER -> EchoLocate.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

GOLD

Master-series microphone voicing that combines source-array blend with silver air, copper body, iron weight, gold presence, platinum authority, and nickel focus.

Declared product: GOLD | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
sourcearray	Source / Mic Array	Factory
AG_AIR	Ag Air	Factory
CU_BODY	Cu Body	Factory
FE_GLUE	Fe Weight	Factory
AU_DRIVE	Au Presence	Factory
PT_AUTHORITY	Pt Authority	Factory
NI_FOCUS	Ni Focus	Factory

Artwork authority

assets/gold_bg.png

SILVER

Precision microphone voicing for truth, focus, grounding, air, and transient definition with protected input, mix, and output control.

Declared product: SILVER | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool
AG_TRUTH	Truth Field	Factory
NI_FOCUS	Precision	Factory
CU_GROUND	Ground	Factory
AIR	Air	Factory
TRANSIENT	Transient	Factory
OUTPUT	Output	Factory
INPUT_GAIN	Input	Factory
MIX_BLEND	Mix	Factory

Artwork authority

assets/silver_bg.png

WPR

Phi ribbon microphone channel with transformer family, core material, winding, lamination, polar pattern, ribbon mass, proximity, and magnetic behavior.

Declared product: WPR | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
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Parameter surface

ID	Display name	Type
DRIVE	Transformer	Float
ARCHITECTURE	Ribbon Tone	Float
DEPTH	Ribbon Mass	Float
WIDTH	Side Width	Float
FOCUS	Polar Focus	Float
ENERGY	Output Trim	Float
CLARITY	Air Edge	Float
MIX	Ribbon Blend	Float
CONVERGENCE	Proximity	Float
IRON_DRIVE	Iron Drive	Float
FLUX	Magnetic Flux	Float
HEADROOM	Core Headroom	Float
PRI_BIAS	Primary Bias	Float
SEC_COLOR	Sec Coloring	Float
LOAD	Winding Load	Float
PERMEABILITY	Permeability	Float
HYSTERESIS	Hysteresis	Float
GEOMETRY	Iron Geometry	Float
gainmatch	Gain Match	Bool
LAMINATION	Lamination Type	Choice
TRANSFORMER_FAMILY	Transformer Family	Choice
CORE_MATERIAL	Core Material	Choice
WINDING_TOPOLOGY	Winding Topology	Choice
CORE_AGE	Core Age	Choice
POLAR_MODE	Polar Pattern	Choice
OUTPUT	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

source/WPR.png

Aegis Nova

Vocal shield for plosive and sibilance control with a restrained drive stage and parallel mix.

Declared product: AEGIS_NOVA | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
gainmatch	Gain Match	Bool
power	Power	Bool
plosive	Plosive	Float
sibilance	Sibilance	Float
drive	Drive	Float
mix	Mix	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

assets/aegisNova.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT	Incoming peak activity	Confirms source level reaching the shield.
NOVA	Detected plosive/sibilant activity	More light means stronger shield intervention.

TRIPLER

Three-voice vocal harmonizer with key, scale, chord, pitch correction, voice gain, pan, fine tune, and Phi field control. The current left/right voices use the repaired high-quality overlap-add pitch path for clean stereo harmonies.

Declared product: TRIPLER | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call not detected; integration review required before release claims
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
DIATONIC	DIATONIC	Referenced
CHORD	CHORD	Referenced
KEY	KEY	Referenced
SCALE	SCALE	Referenced
L_GAIN	L_GAIN	Referenced
R_GAIN	R_GAIN	Referenced
C_GAIN	C_GAIN	Referenced
L_PAN	L_PAN	Referenced
R_PAN	R_PAN	Referenced
BYPASS	BYPASS	Referenced
PITCH_CORRECT	PITCH_CORRECT	Referenced
PC_AMOUNT	PC_AMOUNT	Referenced
FINE_TUNE	FINE_TUNE	Referenced
PHI_FIELD	PHI_FIELD	Referenced
L_ON	L_ON	Referenced
R_ON	R_ON	Referenced
OUT_GAIN	OUT_GAIN	Referenced
GAINMATCH	GAINMATCH	Referenced

Artwork authority

assets/TRIPLER.png

The repaired left/right pitch path uses OSAMP 8, overlap-add normalization, and fractional spectral-bin interpolation to remove the prior light distortion while preserving the three-voice design.

EchoLocate

Three-position source-placement processor for spread, drift, microphone gain, proximity, pan, and reflected-space mix.

Declared product: EchoLocate | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Insert Safe | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

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Parameter surface

ID	Display name	Type
MODE	MODE	Referenced
SPREAD	SPREAD	Referenced
DRIFT	DRIFT	Referenced
MIX	MIX	Referenced
MIC1_GAIN	MIC1_GAIN	Referenced
MIC2_GAIN	MIC2_GAIN	Referenced
MIC3_GAIN	MIC3_GAIN	Referenced
MIC2_PROX	MIC2_PROX	Referenced
MIC3_PAN	MIC3_PAN	Referenced
OUT_GAIN	OUT_GAIN	Referenced
channelmode	channelmode	Referenced
dualmonolink	dualmonolink	Referenced

Artwork authority

resources/echolocate.png